Vatsa Arvind Kala

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Education

Texas A&M University, College Station, TX **Aug 2024 - May 2026**
*Master of Science in Management Information Systems — **GPA: 3.785/4.0*** College Station, Texas

Pandit Deendayal Energy University **Oct 2020 - May 2024**
*B.Tech in Electronics And Communication Engineering — **GPA: 9.42/10*** Gandhinagar, India

Technical skills

Languages & Frameworks: Python, SQL, R, C++, C, JavaScript, Typescript, Kotlin, Java, Flask, React, PyTorch, Node.js
Tools & Tech Stack: AWS (EC2, Fargate, S3, RDS, Redshift, QuickSight), dbt, Tableau, Power BI, Git, Jira, Postman, Kubernetes
Databases & Interests: Snowflake, PostgreSQL, MongoDB, LangChain, LangGraph, Machine Learning, Automation Testing
Certifications: Professional Scrum Master 1, Fundamentals of Deep Learning, Microsoft Azure Data Engineer Associate

Professional experience

Amazon **Sep 2026 - Present**
Software Development Engineer Austin, Texas

- Automated **multiple end-to-end test scenarios** in **Cucumber.js** as owner of the **Fire TV channel manager**, building reusable components across **2 operating systems** to cover all **user entry points** into a business-critical workflow
- Elevated **reliability** of the channel manager by debugging intermittent failures with device logs, input timing, and UI snapshots, and **separating product defects from environment issues** and partnering with owning teams to resolve them
- Developed clean, maintainable **TypeScript** code and technical documentation for the channel manager feasture in an **agile team**, contributing to design discussions and code reviews and **owning changes from design through deployment**

Trading Technologies **May 2025 - Dec 2025**
Data Engineering Intern New York, USA

- Collaborated on **migrating 4+ petabytes** from Microsoft SQL Server to PostgreSQL on AWS RDS and Snowflake, redesigning 500+ tables through renames, primary key changes, and corrected dependency order for cutover
- Engineered a Flask and SQLAlchemy analytics API with 30+ endpoints for the new TradeZoom platform, standardizing data access into reusable query modules and **improving release stability** during migration to cloud data stores
- Conducted migration validation using row counts, null checks, uniqueness checks, and relationship rules, **uncovering about 100+ legacy defects** such as duplicates, missing primary keys, and broken table dependencies before cutover

Texas A&M University **Jan 2025 - May 2025**
Student Assistant College Station, USA

- Elevated transplant survival prediction by packaging an **end-to-end DeepSurv training and evaluation pipeline** on the UNOS thoracic registry, modeling 25,953 lung-only cases and achieving 0.89 test C-index with 0.24 Brier score at 4000 days
- Restructured UNOS data from 215,462 patients and 532 variables into a 73-feature dataset by writing preprocessing and encoding scripts and applying MICE Random Forest imputation for 78% missingness, **avoiding 45% deletion loss**
- Built model trust and explanation layer by running Kaplan–Meier validation, generating SHAP explanations for top risk drivers, and **integrating LLaMA-3.1 8B to convert predictions into clinician-readable narratives** for decision support

Projects

StudyGraph: Agentic AI Learning Pathway Engine | *LangGraph, LangChain, RAG, HuggingFace, FastAPI, React*

- Integrated an agentic AI backend with retrieval, ranking, validation, and adaptive study sequencing to generate personalized pathways **across 4 depth policies**, improving roadmap coverage by **18%** through checkpoints and skip guidance
- Architected a LangGraph state workflow across **14 pipeline nodes** to manage validation, resource retrieval, enrichment, coverage backfill, LLM verification, ranking, and plan critique, reducing API lookups by **26%** through cached responses

Point of sales database system | *Amazon Web Services, MongoDB, ETL, Cloud Architecture*

- Designed a POS relational database on AWS EC2 using MySQL/MariaDB, modeling 10K+ customers, 8K products, and 30K orders; implemented transactions and triggers to **enforce data integrity** across checkout and inventory workflows
- Transformed performance by building ETL using SQL and Skyvia, profiling join bottlenecks, then migrating to MongoDB with a new document model and indexes which improved load 25%, queries 65%, and **supported 50% more transactions**

Custom MCP Neighborhood Server | *Flask, Claude 3.5 Sonnet, REST APIs, OpenAI MCP SDK*

- Constructed a Flask-based MCP neighborhood intelligence server using the OpenAI MCP SDK, **calling 10+ public APIs** including housing, crime and converting unstructured JSON into a consistent feature dataset for address-level analysis
- Orchestrated asynchronous API fan-out with retries, timeouts, and strict schema validation to normalize upstream payloads, **cutting end-to-end response latency by 24%** and increasing response consistency across sources